

High-level components & data flows

- Microsoft Entra ID (IdP): Hosts the enterprise app (AWS IAM Identity Center). Responsible for user authentication (SAML) and can push users/groups to AWS via SCIM provisioning. [AWS Document...](#)
- AWS IAM Identity Center (Service Provider / SP): Consumes IdP SAML metadata, validates SAML assertions, maps users to local IAM Identity Center users and permission sets. Also exposes/configures SCIM provisioning endpoint to accept pushes from Entra. [AWS Document...](#)
- SAML flow: User tries to access AWS portal → IAM IC redirects to Entra SSO URL → Entra authenticates user and sends SAML assertion back to IAM IC → IAM IC maps user & grants roles/permission sets. [AWS Document...](#)
- SCIM flow: Entra provisioning service uses SCIM v2 to create/update users and groups in IAM Identity Center (uses a SCIM base URL + Bearer token). This automates user/group lifecycle. [AWS Document...](#)

3) Detailed step-by-step plan (mapped to the AWS doc)

Follow these steps in order. I include example values and the precise actions the UI requires.

Prereqs

1. Microsoft Entra tenant (Azure AD) with admin rights (Cloud Application Admin or Global Admin). [AWS Document...](#)
2. AWS account with IAM Identity Center enabled (or IAM Identity Center already enabled for the AWS organization). [AWS Document...](#)

Step 1 — Prepare Microsoft Entra

1. In Entra admin center → Identity > Applications > Enterprise applications → New application. Search for AWS IAM Identity Center and create it. [AWS Document...](#)
2. Create a test user (e.g., `NikkiWolf@yourdomain`) under Identity > Users > All users. Note the UPN. [AWS Document...](#)
3. Assign that user to the AWS IAM Identity Center enterprise app (Users & groups → Add user/group → select your test user → Assign). [AWS Document...](#)

Step 2 — Prepare AWS (IAM Identity Center)

1. In AWS console → IAM Identity Center → Settings → Identity source → Change identity source → choose External identity provider and Next. [AWS Document...](#)
2. On that wizard under Service provider metadata choose Download metadata file — save the XML. Also copy the AWS access portal sign-in URL shown on that page (you'll paste it into Entra SAML config). [AWS Document...](#)
3. Create any test permission sets (e.g., `RegionalAdmin`) that will be assigned to users in IAM IC. Permission sets contain the AWS permissions (managed policies or inline). [AWS Document...](#)
4. (Optional) Create a local IAM Identity Center user for initial test mirroring (since SAML alone doesn't create users in IAM IC). Use the same UPN/email as the Entra user (`NikkiWolf@...`). [AWS Document...](#)

Step 3 — Configure SAML in Entra (IdP config)

1. In Entra → open the AWS IAM Identity Center enterprise app → 2. Set up Single sign-on → choose SAML.
2. On Set up SAML, Upload metadata file you downloaded from AWS (service provider metadata). This will populate Identifier and Reply URL automatically. [AWS Document...](#)
3. Set the Sign on URL (optional) to the AWS access portal sign-in URL (copied earlier). Save. Download Entra's Federation Metadata XML (IdP metadata) or get the SAML metadata URL. [AWS Document...](#)

Step 4 — Configure IdP in AWS (complete change identity source)

1. Return to the IAM Identity Center Change identity source wizard in AWS. Upload the IdP SAML metadata (the Federation Metadata XML you downloaded from Entra). Continue the wizard. Enter `ACCEPT` when prompted and apply the change. This completes the SAML trust (AWS becomes SP to Entra). [AWS Document...](#)

Step 5 — Configure SCIM provisioning (automatic user/group sync)

1. In IAM Identity Center, go to Settings → Identity source (or provisioning section) and find the SCIM provisioning information (the documentation shows here where to enable/collect the SCIM endpoint and token that AWS expects). [AWS Document...](#)
2. In Entra enterprise app → Provisioning tab → set Provisioning Mode to Automatic and fill Tenant URL (SCIM Base URL) and Secret Token (Bearer token) copied from AWS IAM Identity Center. Save and

test the connection.

- Note: Microsoft Entra will ask for a Client Secret only when you register a custom app — with the AWS gallery app, Entra expects a SCIM endpoint URL and secret token values. The AWS doc covers how to retrieve the SCIM URL/token from IAM Identity Center. [AWS Document...](#)

3. Configure scoping / attribute mapping in Entra (which groups/users to sync). Use assignment-based scope or dynamic groups depending on your design. Test by provisioning a test user (e.g., RichardRoe in doc) and confirm it appears in IAM Identity Center. [AWS Document...](#)

Step 6 — Testing SSO and Provisioning

1. Test SAML sign-on: From AWS access portal URL → choose Sign in → you should be redirected to Entra and return with a SAML assertion — confirm user created/mapped and assigned permission set grants expected access. [AWS Document...](#)
2. Test SCIM: Create or update user/group in Entra (or assign to the AWS app) and confirm the user is created/updated in IAM Identity Center. Test group membership propagation. [AWS Document...](#)

4) Example values & SAML attribute mapping (practical)

- Example AWS sign-in URL (portal): `https://<region>.signin.aws.amazon.com/console` (the exact AWS access portal URL is shown in the IAM IC settings). [AWS Document...](#)
- Example SAML attributes to provide from Entra → AWS (common):
 - NameID → user principal name (UPN) / email
 - `https://aws.amazon.com/SAML/Attributes/Role` → `arn:aws:iam::<AWS_ACCOUNT_ID>:role/<RoleName>,arn:aws:iam::<AWS_ACCOUNT_ID>:saml-provider/<ProviderName>` (if doing role chaining in OIDC/SAML flows)
 - `https://aws.amazon.com/SAML/Attributes/RoleSessionName` → username/email
 - In many IAM IC setups you do not need to handcraft roles because IAM Identity Center maps SAML users to local permission sets. Confirm with your permission model. [AWS Document...](#)

5) Security & operational notes

- Least privilege: permission sets should follow least privilege and use session durations compatible with your security posture. [AWS Document...](#)
- SCIM limits: Entra provisioning has limitations (e.g., nested groups and dynamic group caveats). If you rely on nested groups, plan alternative provisioning or flatten groups. [AWS Document...](#)
- Attribute removal limitation: If an attribute is removed in Entra, it may not be removed in IAM Identity Center automatically — design workflows for attribute cleanup. [AWS Document...](#)
- Testing: Use isolated test accounts and test permission sets before production cutover. Create a rollback plan (revert identity source) before making changes in production.

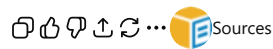
6) Troubleshooting quick checklist

- SAML assertion fails: ensure Audience / Identifier and Reply URLs match values in both metadata files. [AWS Document...](#)
- SCIM provisioning fails: verify SCIM base URL and secret token copied from IAM IC are correct and provisioning is enabled in Entra. Also check Entra provisioning logs for HTTP status and errors. [AWS Document...](#)
- Group sync missing members: check nested group handling / dynamic-group limitations in Entra. [AWS Document...](#)

If you want, I can *also*:

- produce a PNG (rendered diagram) and a one-page PDF with the diagram + step list, or
- generate a Visio (.vsdx) or SVG file for your architecture diagram, or
- create a ready-to-run terraform outline to automate parts of the setup (e.g., IAM Identity Center permission sets + SAML provider resources).

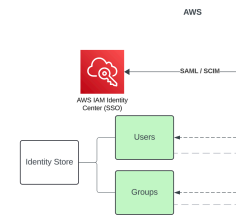
Which of those would you like me to generate now — PNG, PDF, Visio, or Terraform scaffolding?



Sources

generate me PNG and PDF and step list

Cool, I found a diagram that's pretty close in style. I can't generate a PNG or PDF file *directly here*, but I can give you what you need to plug into free tools to produce both (or send to a designer). Meanwhile, here's a refined step list + guidance for exporting.



Refined Step List

Use this list as your checklist when implementing the architecture (SAML IdP + SCIM provisioning between Microsoft Entra ID and AWS IAM Identity Center):

Step	Action	Key Inputs / Checks
1. Pre-setup	Ensure permissions and accounts are ready	You must have admin privileges on both Mi (Azure AD) and AWS with IAM Identity Cent regions.
2. Create Entra Enterprise App	In Microsoft Entra ID → <i>Enterprise Applications</i> → <i>New application</i> → choose AWS IAM Identity Center*	Give it a name like "AWS-IAM-IC-SSO/Provi
3. Add test user/group in Entra	Create test user(s); create or identify group(s) for AWS access; assign them to the AWS app	Record UPN / email; ensure correct group r
4. In AWS IAM Identity Center: Change identity source	Go to IAM Identity Center -> Settings -> Identity Source -> change to External Identity Provider*	Download SP metadata, grab the AWS acce
5. In Entra: Set up SAML in Enterprise App	Upload SP metadata from AWS; configure Identifier / Reply URLs; set Sign on URL; optionally configure attribute mappings and certificates	Ensure values match exactly.
6. In AWS: Upload IdP metadata	In the IAM Identity Center identity source setup, upload the Federation Metadata XML from Entra	Complete the trust; confirm identity source
7. Configure SCIM provisioning	In AWS, get SCIM endpoint URL + secret token; in Entra app → <i>Provisioning</i> → set to automatic + these values	Test connection; decide scope (which users,
8. Assign groups/users in Entra to the Enterprise App	So that SCIM will push them; ensure they have the correct attributes	Check for correct attributes (email, name, g membership).
9. Create Permission Sets in AWS	Define what access users/groups should have	Examples: S3 read, EC2 admin, etc. Use leas
10. Assign Permission Sets to Users/Groups	Within IAM Identity Center assign to the groups imported via SCIM or test users	Verify roles are created in AWS accounts.
11. Test SSO login	User tries to access AWS portal; gets redirected to Entra; after login lands in AWS with expected permissions	Check that roles are correct; session works.
12. Test SCIM provisioning	Add a user in Entra → see them show up in IAM IC; remove/update group membership → changes reflected	Review provisioning logs if failures.
13. Audit & Security Review	Review certificate lifetimes, token security, session duration; ensure compliance	Set up monitoring.
14. Rollout & Documentation	Train users/admins; document the configuration; plan fallback in case identity source issues	Keep backups of metadata files, secrets sto